


Linear Algebra

# Bases & Dimension

Department of Computer Engineering

Sharif University of Technology

Hamid R. Rabiee rabiee@sharif.edu
Maryam Ramezani maryam.ramezani@sharif.edu

# Table of contents


01 Introduction

02 Basis

03 Dimension

04 Finite Dimensional Subspace

05 Coordinates

# 01

# Introduction

# Price Problem



$ 70’000

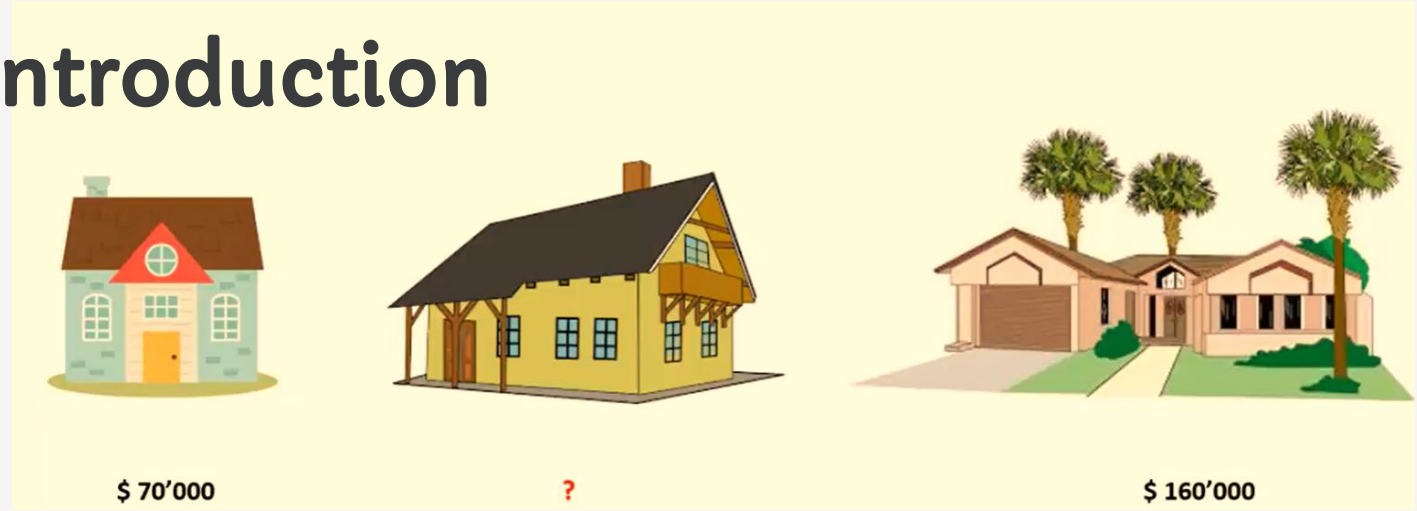


?



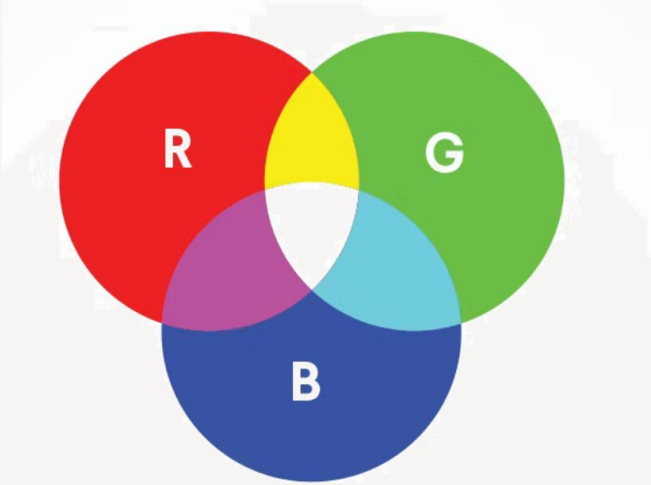
$ 160’000

# Introduction




$$
\begin{array}{l}
\\
\text{Home \#1} \\
\text{Home \#2} \\
\text{Home \#3} \\
\\
\text{Home \#N}
\end{array}
\begin{bmatrix}
\#Room & Size_part1 & Size_part2 & Size_part3 & Size_part4 & Size & Age & Floor & Is_near_park \\
 & & & & & & & & \\
 & & & & & & & & \\
 & & & & & & & & \\
 & & & & & & & & \\
 & & & & & & & &
\end{bmatrix}
$$

# Introduction






# 02

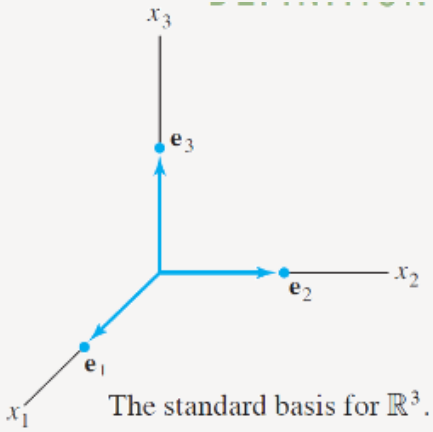
# Basis

# Basis

- A set of n linearly independent n-vectors is called a basis.
- A basis is the combination of span and independence: A set of vectors $\{v_1, \ldots, v_n\}$ forms a basis for some subspace of $\mathbb{R}^n$ if it

  (1) spans that subspace

  (2) is an independent set of vectors.



The standard basis for $\mathbb{R}^3$.

# Basis

**Definition**

Let $H$ be a subspace of a vector space $V$. An indexed set of vectors $\mathcal{B} = \{b_1, \dots, b_n\}$ in $V$ is a **basis** for $H$ if

1. $\mathcal{B}$ is linearly independent set, and
2. The subspace spanned by $\mathcal{B}$ coincides with $H$; that is,

$$H = Span\,\{b_1, \dots, b_n\}$$

**Example**

Which are unique?

- ❑ Express a vector in terms of any particular basis
- ❑ Bases for $\mathbb{R}^2$
- ❑ Bases with unit length for $\mathbb{R}^2$

# Example Solution!

- **Express a vector in terms of any particular basis**
  **Unique:** Once a basis is fixed, the coefficients (coordinates) are uniquely determined.

- **Bases for $\mathbb{R}^2$**
  **Not unique:** There are infinitely many possible bases, as any two linearly independent vectors can form one.

- **Bases with unit length for $\mathbb{R}^2$**
  **Not unique:** There are infinitely many orthonormal bases (rotations of each other).

# Vector Space of Polynomials

**Be careful:** A vector space can have many bases that look very different from each other!

Example (Basis)

- Standard bases for $P_n(\mathbb{R})$?
- Are $(1-x), (1+x), x^2$ basis for $P_2(\mathbb{R})$?

# Example Solution!

- **Standard basis for $P_n(\mathbb{R})$:**
  The standard basis is $\{1, x, x^2, \dots, x^n\}$.
  Each polynomial in $P_n(\mathbb{R})$ can be uniquely written as a linear combination of these monomials.

- **Are (1−x), (1+x), x² a basis for $P_2(\mathbb{R})$?**
  Yes. They are **linearly independent**.
  $P_2(\mathbb{R})$ is **spanned** by these three vectors.
  Therefore, they form a valid basis for $P_2(\mathbb{R})$.

# 03

# Dimension

# Dimensions

- The dimensionality of a vector is the number of coordinate axes in which that vector exists.
- If a vector space is spanned by a finite number of vectors, it is said to be finite-dimensional. Otherwise, it is infinite-dimensional.
- The number of vectors in a basis for a finite-dimensional vector space V is called the dimension of V and denoted dim(V).

# Bases and finite dimension

## Theorem 1

Let $V$ be a vector space which is spanned by a finite set of vectors $x_1, x_2, \dots, x_m$. Then any independent set of vectors in $V$ is finite and contains no more than $m$ elements. Independent ≤ spanning

## Conclusion

Every basis of $V$ is finite and contains no more than $m$ elements.

# Proof of Theorem 1

Because $\{x_1, \dots, x_m\}$ spans $V$, every vector in $V$ can be written as a linear combination of these $m$ vectors.

Now suppose we have a linearly independent set $\{y_1, \dots, y_n\}$ in $V$ with $n > m$.

Each $y_i$ can be expressed as

$$y_i = a_{i1}x_1 + \cdots + a_{im}x_m$$

Consider a linear combination of the $y_i$'s that equals the zero vector:

$$0 = c_1y_1 + \cdots + c_ny_n$$

Substitute each $y_i$ in terms of the $x_j$'s:

$$(c_1a_{11} + \cdots + c_na_{n1})x_1 + \cdots + (c_1a_{1m} + \cdots + c_na_{nm})\,x_m = 0$$

Since the $x_j$'s span $V$, this gives a homogeneous system of $m$ equations in $n$ unknowns $c_1, \dots, c_n$. If $n > m$, this system has a nontrivial solution (some $c_i \neq 0$), which means the $y_i$'s are linearly dependent. ⇒ Contradiction!

# Bases and finite dimension

## Theorem 2

If $V$ is a finite-dimensional vector space, then any two bases of $V$ has the same (finite) number of elements.

## Proof of Theorem 2

Let $B_1 = \{x_1, \dots, x_m\}$ and $B_2 = \{y_1, \dots, y_n\}$ be two bases of $V$.
Since $B_1$ spans $V$, any independent set (like $B_2$ ) can have **no more than** $m$ elements $\Rightarrow n \leq m$.
Likewise, since $B_2$ spans $V$, any independent set (like $B_1$) can have **no more than** $n$ elements $\Rightarrow m \leq n$.
Hence $m = n$.

# Basis and finite dimension

The number of vectors in a basis for a finite-dimensional vector space V is called the dimension of V and denoted as dim(V).

**Theorem 2**



## Theorem 3

Let $V$ be a vector space with a basis $B$ of size $m$. Then

a) Any set of more than $m$ vectors in $V$ must be linearly dependent, and

b) Any set of fewer than $m$ vectors cannot span $V$.

# Proof of Theorem 3

a) Since $B$ is a basis, it spans $V$. By theorem 2, any linearly independent set in $V$ can have at most $m$ elements. Therefore, any set with more than $m$ vectors must be linearly dependent.

b) Suppose, for contradiction, that there exists a spanning set $\{y_1, \dots, y_k\}$ with $k < m$. From $S$, we can extract a basis (by removing dependent vectors). That basis would have at most $k$ elements. But this contradicts the fact that every basis of $V$ must have $m$ elements.

# Dimensions

## Definition

A vector space $V$ is called...

a) **finite-dimensional** if it has a finite basis, and its **dimension**, denoted by $\dim(V)$, is the number of vectors in one of its bases.

b) **infinite-dimensional** if it has no finite basis, and we say that $\dim(V) = \infty$.

## Note

Dimension of subspace $\{\mathbf{0}\}$?

# 04

# Finite Dimensional Subspace

# Basis of Subspace

## Theorem 4

If $W$ is a subspace of a finite-dimensional vector space $V$, every linearly independent subset of $W$ is finite and is part of a (finite) basis for $W$.

## Lemma

Let $S$ be a linearly independent subset of a vector space $V$. Suppose $u$ is a vector in $V$ which is not in the subspace spanned by $S$. Then the set obtained by adjoining $u$ to $S$ is linearly independent.

# Proof of Lemma

Assume

$$a_1v_1 + \cdots + a_nv_n + cu = 0$$

If $c \neq 0$, we can solve for $u$: $u = -\frac{1}{c}(a_1v_1 + \cdots + a_nv_n)$

This means $u$ is a linear combination of vectors in $S$, i.e., $u \in \text{span}(S)$, which contradicts our assumption that $u \notin \text{span}(S)$.

Therefore, $c = 0$. The equation becomes $a_1v_1 + a_2v_2 + \cdots + a_nv_n = 0$.

Since $S$ is linearly independent, this implies $a_1 = a_2 = \cdots = a_n = 0$.

Thus, all coefficients are zero, so $S \cup \{u\}$ is linearly independent.

# Proof of Theorem 4

Let $\dim V = m$. Every linearly independent set in $V$ has at most $m$ elements.

Hence any linearly independent subset of $W$ also has size $\leq m$, so it is finite.

Therefore, the possible sizes of independent subsets of $W$ lie in the finite set $\{0,1,\dots,m\}$.

So largest possible size exists. Choose an independent subset $B \subseteq W$ that has this maximal size.

We claim B spans W. If not there exits a $w \notin span(B)$.

By lemma, $B \cup \{w\}$ must be linearly independent, contradicting maximality of B. Thus, B is a finite basis for W.

# Basis of Subspace

## Corollary

A subspace is called a proper subspace **if it's not the entire space**, so R2 is the only subspace of R2 which is not a proper subspace

If $W$ is a proper subspace of a finite-dimensional vector space $V$, then $W$ is finite-dimensional and $\dim(W) < \dim(V)$

## Corollary

In a finite-dimensional vector space $V$, every non-empty linearly independent set of vectors is part of basis.

# Basis of sum of subspaces

## Theorem 5

If $W_1$ and $W_2$ are finite-dimensional subspaces of a vector space $V$,

the $W_1 + W_2$ is a finite-dimensional and

$$\dim(W_1) + \dim(W_2) = \dim(W_1 \cap W_2) + \dim(W_1 + W_2)$$

# Proof of Theorem 5

Let $v_1, \dots, v_m$ be a basis of $W_1 \cap W_2$, so $\dim(W_1 \cap W_2) = m$.

Extend this basis to a basis of $W_1$ : $v_1, \dots, v_m, u_1, \dots, u_j$ ,so $\dim(W_1) = m + j$.

Similarly, extend $v_1, \dots, v_m$ to a basis of $W_2$: $v_1, \dots, v_m, w_1, \dots, w_k$ ,so $\dim(W_2) = m + k$.

We will show that $v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$ is a basis of $W_1 + W_2$.

**1. Spanning:**

Every vector in $W_1$ is a linear combination of $v$'s and $u$'s, and every vector in $W_2$ is a linear combination of $v$'s and $w$'s. Hence the span of all these vectors contains both $W_1$ and $W_2$, so it equals $W_1 + W_2$.

# Proof of Theorem 5 (cont.)

**2. Linear Independence:**

Suppose $a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j + c_1w_1 + \cdots + c_kw_k = 0$.

Move the $w$-terms to one side:

$-(a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j) = c_1w_1 + \cdots + c_kw_k$.

The right-hand side is in $W_1$, and the left-hand side is in $W_2$;hence this vector is in $W_1 \cap W_2$, which is spanned by $v_1, \ldots, v_m$ .Thus

$d_1v_1 + \cdots + d_mv_m = c_1w_1 + \cdots + c_kw_k$ for some $d_i$'s.

Since $v_1, \ldots, v_m, u_1, \ldots, u_j, w_1, \ldots, w_k$ are linearly independent all $c_i$'s (and $d_i$'s) are 0, so we have: $a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j = 0$ which implies all $a_i$'s and $b_i$'s 0 since these vectors are basis of $W_1$. Hence all vectors are linearly independent.

# Basis of sum of subspaces

## Theorem 6

If $W_1,\ W_2\ and\ W_3$ are finite-dimensional subspaces of a vector space $V$, then can we have the following relation?

$$\dim(W_1 + W_2 + W_3)$$
$$= \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2)$$
$$- \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) + \dim(W_1 \cap W_2 \cap W_3)$$

**Counterexample:** $W_1 = span\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}, W_2 = span\{\begin{bmatrix}0\\1\end{bmatrix}\}, W_3 = span\{\begin{bmatrix}1\\1\end{bmatrix}\}$

# Basis of sum of subspaces

## Theorem 6

If $W_1,\ W_2\ and\ W_3$ are finite-dimensional subspaces of a vector space $V$, then:

$$\dim(W_1 + W_2 + W_3)$$
$$\leq \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2)$$
$$- \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) + \dim(W_1 \cap W_2 \cap W_3)$$

**Problem 24:** *Let $W_1$, $W_2$, $W_3$ be subspaces of a F.D.V.S. Show that*

$$\dim (W_1 + W_2 + W_3) \leq \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_1 \cap W_2) - \dim (W_1 \cap W_3) - \dim (W_2 \cap W_3) + \dim (W_1 \cap W_2 \cap W_3).$$

**Solution:** We have

$$\begin{aligned}\dim (W_1 + W_2 + W_3) &= \dim W_1 + \dim (W_2 + W_3) - \dim (W_1 \cap (W_2 + W_3)) \\ &= \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_2 \cap W_3) \\ &\qquad - \dim (W_1 \cap (W_2 + W_3)) \\ &\leq \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_2 \cap W_3) \\ &\quad - \dim (W_1 \cap W_2) - \dim (W_1 \cap W_3) + \dim (W_1 \cap W_2 \cap W_3)\end{aligned}$$

as $(W_1 \cap W_2) + (W_1 \cap W_3) \subseteq W_1 \cap (W_2 + W_3)$.

# Dimensionality and Properties of Bases

## Note

Let $V$ be a finite dimensional vector space over field $F$. Below are some properties of bases:

1. Any linearly independent list can be extended to a basis (a maximal linearly independent list is spanning).
2. Any spanning list contains a basis (a minimal spanning list is linearly independent).
3. Any linearly independent list of length dim $V$ is a basis.
4. Any spanning list of length dim $V$ is a basis.

**We will learn about change of basis later.**

05

# Coordinates

# Ordered basis

## Definition

If $V$ is a finite-dimensional vector space, as ordered basis for $V$ is a finite sequence of vectors which is linearly independent and spaces $V$.

**Be careful:** The order in which the basis vectors appear in $B$ affects the order of the entries in the coordinate vector. This is kind of janky (technically, sets don't care about order), but everyone just sort of accepts it.

# Coordinate Systems

- The main reason for selecting a basis for a subspace $H$; instead of merely a spanning set, is that each vector in $H$ can be written in only one way as a linear combination of the basis vectors.

## Note

Suppose the set $\mathcal{B} = \{\boldsymbol{b_1}, \dots, \boldsymbol{b_P}\}$ is a basis for a subspace $H$. For each $x$ in $H$, the **coordinates of $x$ relative to the basis $\mathcal{B}$** are the weights $c_1, \dots, c_P$ such that $\mathrm{x} = c_1 b_1 + \cdots + c_P b_p$, and the vector in $\mathbb{R}^p$

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_P \end{bmatrix}$$

is called the **coordinate vector of $x$ (relative to $\mathcal{B}$)** or the $\mathcal{B}$-coordinate vector of $x$.

# Coordinate Systems

## Example

Coordinate vector of $p(x) = 4 - x + 3x^2$ respect to basis $\{1, x, x^2\}$

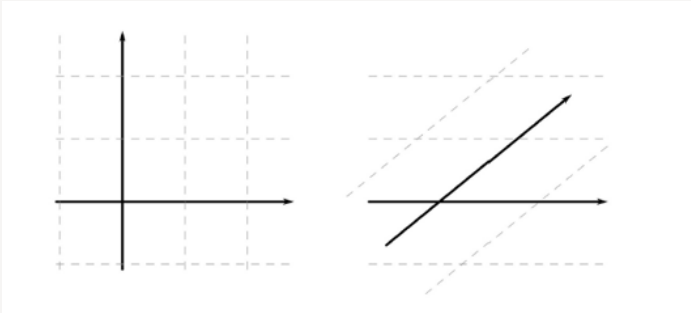
**Example Solution!**

$$p(x) = 4(1) - 1(x) + 3(x^2)$$

$$[p(x)]_B = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}$$

# Coordinate axes

- The familiar Cartesian plane (left) has orthogonal coordinate axes. However, axes in linear algebra are not constrained to be orthogonal (right), and non-orthogonal axes can be advantageous.



# Resources

- Page 97 LINEAR ALGEBRA: Theory, Intuition, Code
- Page 213: David Cherney,
- Page 54: Linear Algebra and Optimization for Machine Learning

# Dimensions

## Example

Let's compute the dimension of some vector spaces that we've been working with.

| Vector space | Basis | Dimension |
|---|---|---|
| $\mathcal{F}^n$ (n–tuples each elements from field $\mathcal{F}$) | | |
| $P^p$ (polynomials with max degree $p$) | | |
| $M_{m,n}$ (matrices with m rows and n columns) | | |
| $P$ (all polynomials) | | |
| $F$ $(all\ functions)$ | | |
| $C$ (all continues functions) | | |

Note!

**Example.** *Let's compute the dimension of some vector spaces that we've been working with.*

| $V$ | Basis | $\dim(V)$ |
|---|---|---|
| $\mathbb{F}^n$ ($\mathbb{R}^n$, $\mathbb{C}^n$, etc.) | $\{\vec{e}_1, \vec{e}_2, \cdots, \vec{e}_n\}$ | $n$ |
| $P^p$ | $\{1, x, x^2, \cdots, x^p\}$ | $p+1$ ← not $p$! |
| $M_{m,n}$ | $\{E_{1,1}, E_{1,2}, \cdots, E_{m,n}\}$ | $mn$ |
| $P$ (all polynomials) | $\{1, x, x^2, x^3, \cdots\}$ | $\infty$ |
| $\mathcal{F}$ | ?? | $\infty$ |
| $\mathcal{C}$ (continuous functions) | ?? | $\infty$ |